

Title: Epoch Rejection

Category: Preprocessing

1. Objective

To improve the quality of the dataset by identifying and discarding "Bad Trials" (Epochs) that contain excessive noise or movement artifacts. Instead of trying to repair massive artifacts—which might distort the data—we simply remove the entire trial to ensure the Machine Learning model trains only on clean, reliable signals.

2. How it Works

The algorithm checks every trial individually against two strict criteria. If a trial fails either check, it is rejected completely.

1. Amplitude Check (The "Spike" Test):

- We look at the absolute maximum value in the trial.
- Rule: If any signal value > 100 μV (the first parameter) (Amplitude Threshold), the trial is rejected. This usually catches eye blinks or jaw clenches.

2. Variance Check (The "Consistency" Test):

- We compare the (variance) of all channels within the trial.
- Rule: If one channel is significantly noisier than the others (High Z-Score), the trial is rejected.

3.Example: (Trial Evaluation) Scenario: Threshold = 100 μV . We have 3 Trials.

Trial ID	Max Amplitude (μV)	Status	Action	Reason
Trial 1	15.4	Pass	Keep	Signal is within normal brain limits.
Trial 2	500.2	Fail	Drop	Exceeds threshold (500 > 100).
Trial 3	120.5	Fail	Drop	Exceeds threshold (120 > 100).

4. Visual Example:

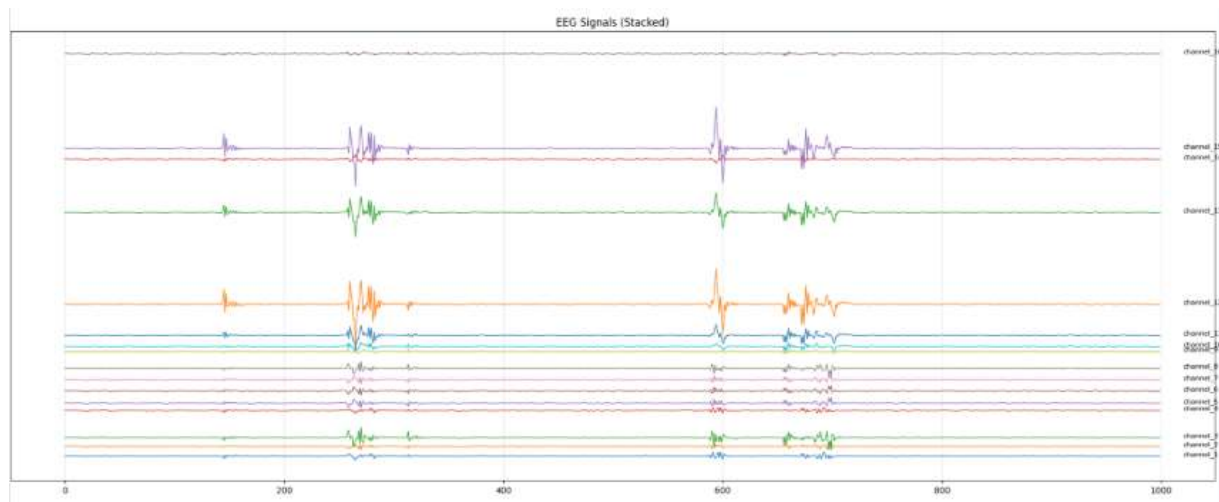


Figure 1 Raw Data

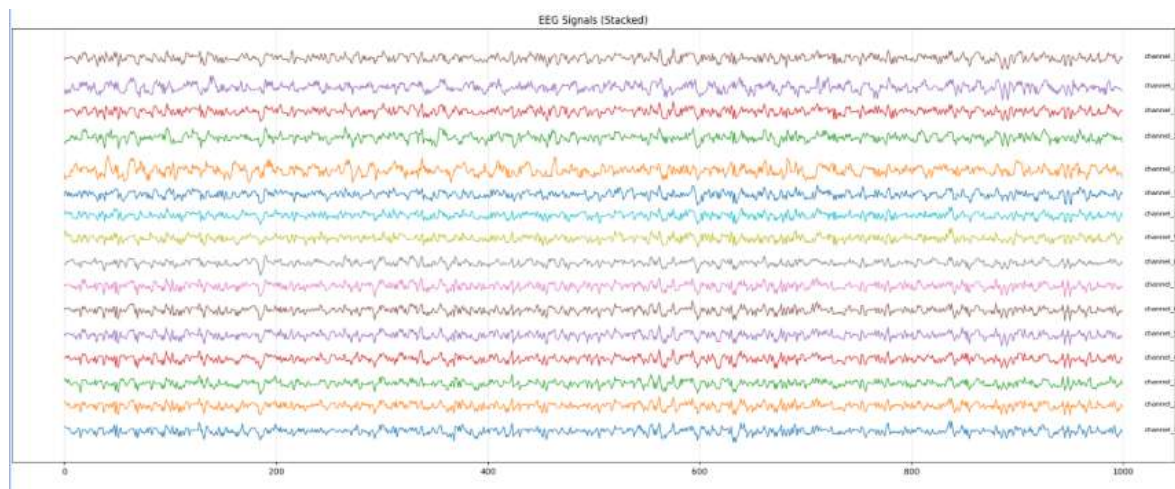


Figure 2 Smoothed Data